HM Government

T-LEVELS

Student Workbook: Temperature and Rate of Reaction

T Level Technical Qualification in Science • Chemistry • A11.2

Name: _____ Date: _____

Specification coverage

Point	Required understanding
A11.2.1	Higher temperature: faster particles, increased kinetic energy, increased collision frequency and greater proportion of successful collisions.
A11.2.2	Lower temperature: decreased collision frequency and smaller proportion of successful collisions.
A11.2.3	Effect of changing temperature in terms of the Maxwell–Boltzmann distribution.

1. Retrieval

1.1 Collision theory

State the two requirements for a successful collision and define activation energy.

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2. Increasing temperature

Quantity	Effect of increasing temperature	Why
Average kinetic energy		
Particle speed		
Collision frequency		

Pearson

Proportion with E ≥ Ea		
Rate of reaction		

2.1 Full explanation

Explain, step by step, why increasing temperature increases reaction rate. Include kinetic energy, collision frequency, activation energy and successful collisions.

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2.2 Orientation check

Does increasing temperature remove the need for correct spatial orientation? Explain.

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3. Decreasing temperature

Quantity	Effect of decreasing temperature	Consequence for rate
Average kinetic energy		
Particle speed		
Collision frequency		
Proportion with E ≥ Ea		
Successful collisions per unit time		

3.1 Low temperature prediction

Explain why lowering temperature decreases the rate of reaction.

4. Maxwell–Boltzmann distributions

4.1 Axes and threshold

State what the x-axis and y-axis represent on a Maxwell–Boltzmann distribution. What does a vertical Ea line represent?

4.2 Higher temperature curve

Describe how a Maxwell–Boltzmann distribution changes when temperature increases.

4.3 Area beyond Ea

Explain what the area to the right of Ea represents and why it is larger at higher temperature.

4.4 What stays constant

For the same uncatalysed reaction, does increasing temperature change Ea? Explain.

5. Distribution interpretation

5.1 Compare curves

Curve A is tall and narrow. Curve B has a lower peak and is broader toward higher energies. Identify the higher-temperature curve and justify.

5.2 Rate link

Use the curves above to explain which sample has the greater reaction rate.

6. Error Detective

6.1

Correct this statement: 'Heating lowers activation energy.'

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6.2

Correct this statement: 'The Maxwell–Boltzmann peak becomes higher when temperature increases.'

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6.3

Correct this statement: 'Only collision frequency changes when temperature increases.'

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6.4

Correct this statement: 'Every collision with energy above Ea must react.'

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7. Exam style application

7.1 [2]

A reaction mixture is heated. Explain the effect on average kinetic energy and collision frequency.

7.2 [2]

Explain why the proportion of successful collisions increases when temperature rises.

7.3 [3]

Describe the change in the Maxwell–Boltzmann distribution at higher temperature.

7.4 [3]

Explain why heating increases reaction rate using collision theory.

8. Final knowledge check

8.1

How does higher temperature affect average kinetic energy?

8.2

How does it affect collision frequency?

8.3

How does it affect the proportion with $E \geq Ea$?

8.4

What happens to the Maxwell–Boltzmann peak at higher temperature?

8.5

What happens to rate when temperature decreases, and why?

8.6

Does temperature alone change Ea for the same reaction?